Technology Stack

Date	1 July 2026
Team ID	xxxxxx
Project Name	Rising Waters: A Machine Learning Approach to Flood Prediction
Maximum Marks	2 Marks

Define Your Technology Stack:

Identify the programming languages, frameworks, databases, front-end tools, back-end tools, and APIs that you will use to build your solution. A well-chosen technology stack ensures that your application is scalable, maintainable, and aligned with your project requirements.

Fill out the table below to detail the specific technologies chosen for each component of your architecture and provide a brief justification for your choice.

Technology Stack Details

S.No	Architecture Component / Layer	Technology Chosen	Justification / Purpose
1	**Frontend / Client-Side** *(e.g., React, Angular, HTML/CSS)*	HTML, CSS, JavaScript	Used to create a responsive and user-friendly interface for displaying flood prediction results and alerts.

2	**Backend / Server-Side** *(e.g., Node.js, Python/Django, Java)*	Python, Flask	Python supports machine learning integration and Flask helps in handling backend logic and API communication efficiently.
3	**Database / Data Storage** *(e.g., MySQL, MongoDB, PostgreSQL)*	MySQL	Stores historical rainfall data, water level records, prediction outputs, and user information securely.
4	**Cloud / Hosting / Deployment** *(e.g., AWS, Azure, Heroku, Docker)*	AWS Cloud	Provides scalable cloud storage, secure hosting, and reliable deployment for continuous monitoring.
5	**Version Control & CI/CD**	GitHub	Helps manage project code, team collaboration, version tracking, and project deployment workflow.

	(e.g., GitHub, GitLab, Jenkins)		
6	**Third-Party APIs / Other Tools** *(e.g., Stripe, Twilio, Google Maps)*	Weather API, Scikit-learn, Pandas, NumPy	Weather API provides live weather data, while machine learning libraries process data and generate flood predictions accurately.